

Proximal Balancing

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We study causal adjustment with observed pretreatment covariates X , one pretreatment proxy W , and an unobserved confounder. In the canonical regime, W supervises a finite representation $Z = \phi(X)$ but is not included in the final adjustment variable. We establish an exact bias decomposition and show that, under a uniform outcome-relevant stability condition, the observable residual imbalance $D(\phi)$ yields the sensitivity certificate $|\tau_\phi - \tau| \leq \Gamma D(\phi)$. Exact balance identifies the average treatment effect under the stated proxy conditions, while a finite-class learning analysis controls the certificate achieved by empirical representation selection. We separately study a proxy-augmented audit-view construction that prevents literal self-conditioning; its point-identification result requires additional multi-view structure and remains a conditional extension. Accordingly, we do not claim that one generic proxy alone point-identifies the average treatment effect.

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1. Introduction

1.1. Related Works

Balancing scores and covariate balancing. The propensity score is the canonical reduction of observed pretreatment covariates: under treatment ignorability given X , conditioning on $e(X) = P(A = 1 | X)$ balances the distribution of X across treatment arms and identifies causal effects (Rosenbaum and Rubin, 1983). Entropy balancing and the covariate-balancing propensity score instead choose weights or treatment models that target balance directly (Hainmueller, 2012; Imai and Ratkovic, 2014). These methods motivate our optimization target, but their usual causal guarantee still requires the measured covariates to suffice for confounding control. Our setting keeps a latent confounder and asks what an observed proxy can certify about the bias remaining after a learned reduction of X .

Causal representation learning. Representation-learning methods compress high-dimensional covariates while penalizing imbalance between treated and control representations. For example, counterfactual regression bounds counterfactual prediction error using factual prediction error and an integral probability metric between learned treatment-arm representations (Shalit et al., 2017). Our construction shares the use of a learned low-dimensional representation and a distributional balance penalty. The difference is the role of the proxy: W is not an outcome predictor in the canonical method, but an outcome-free supervision signal for choosing $Z = \phi(X)$ when ignorability given X may fail.

Negative controls and proximal causal inference. Negative-control variables can reveal residual confounding because they should not be causally affected by the treatment or should not causally affect the outcome, while remaining associated with hidden common causes (Lipsitch et al., 2010). Proximal causal inference turns this diagnostic idea into point identification by assigning distinct treatment-inducing and outcome-inducing proxy roles and imposing completeness and bridge conditions (Miao et al., 2018; Tchetgen Tchetgen et al., 2024); semiparametric proximal theory then develops influence functions and efficient estimators for the resulting identifying functionals (Cui et al., 2024). Our canonical result has a different target: it uses one pretreatment proxy to bound the bias of adjustment for $Z = \phi(X)$, and it treats an outcome bridge as a structural witness for a sensitivity budget rather than as an estimated nuisance function.

Single-proxy identification. Single Proxy Control shows that one negative-control outcome can identify the effect of treatment on the treated under a counterfactual proxy condition and suitable bridge structure (Park et al., 2024). This differs from our population ATE target, from our latent-confounder proxy interpretation, and from our use of an observable balance discrepancy. The comparison is nevertheless central: it makes clear that “one proxy” does not by itself identify a common estimand. Identification depends on what the proxy measures, which conditional independence is imposed, and whether the analysis uses a bridge, an adjustment score, or an independent audit view.

Kernel discrepancies and sensitivity certification. Maximum mean discrepancy is an integral probability metric that equals zero for identical distributions when its kernel is characteristic (Gretton et al., 2012). We use conditional treatment-arm MMDs of (X, W) as observable magnitudes, not kernel-test p -values. The new theoretical object is the product of this discrepancy and an outcome-relevant stability constant, which yields a bias certificate and a representation-selection guarantee. The individual ingredients have close precedents in balancing, representation learning,

kernel testing, and proximal inference; the manuscript therefore makes no priority claim before a systematic literature audit, and instead states its contribution through the exact assumptions, estimand, and guarantee proved below.

2. Problem Setting

2.1. Preliminaries

Structural causal models. A structural causal model (SCM) consists of endogenous variables, exogenous disturbances, and structural assignments. Writing the endogenous variables as $V_1, \dots, V_d$, an acyclic SCM has equations

$$V_j = f_j(\text{pa}_j, E_j), \quad j = 1, \dots, d, \quad (1)$$

where pa_j contains the direct causes of V_j and E_j is an exogenous disturbance. The intervention $\text{do}(A = a)$ replaces the structural equation for A by the constant a while leaving the other equations unchanged. The resulting outcome is the potential outcome $Y(a)$. An average causal effect is *identified* when its value is uniquely determined by the observed-data distribution for every SCM in the stated model class (Pearl, 2009).

Proximal causal inference. Ordinary adjustment seeks observed covariates that block all non-causal paths from treatment to outcome. Proximal causal inference instead allows an unobserved confounder and uses proxy measurements of it. Standard proximal formulations assign two different proxy roles and posit a bridge equation whose solution translates an observable conditional moment into the target causal functional (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). Completeness or injectivity prevents distinct latent laws from producing the same observable proxy law. These conditions do not follow from the mere existence of a correlated proxy. Our method retains this distinction: injectivity is used for exact interpretation, while quantitative stability is used when balance is only approximate.

2.2. Problem Setting

We observe independent copies

$$O_i = (X_i, W_i, A_i, Y_i), \quad i = 1, \dots, n, \quad (2)$$

where X is a vector of observed pretreatment covariates, W is one observed pretreatment proxy, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome. The variable U denotes an unobserved pretreatment cause that may affect both A and Y . The target is the population average treatment effect

$$\tau \triangleq \mathbb{E}\{Y(1) - Y(0)\}. \quad (3)$$

We do not assume that $A \perp\!\!\!\perp U \mid X$.

The canonical method fixes an integer $M \geq 1$, writes $[M] \triangleq \{1, \dots, M\}$, and prespecifies a class Φ

of measurable maps from the support of X to $[M]$. For $\phi \in \Phi$, the adjustment representation is

$$Z \triangleq \phi(X). \quad (4)$$

This is a construction, not an assumption. The proxy W supervises the choice of ϕ through an outcome-free balance criterion, but W is not included in the canonical adjustment variable Z .

For a fixed ϕ , define the population contrast obtained by adjusting for Z as

$$\tau_\phi \triangleq \sum_{z=1}^M P(Z = z) [\mathbb{E}(Y \mid A = 1, Z = z) - \mathbb{E}(Y \mid A = 0, Z = z)]. \quad (5)$$

The central questions are whether observable balance can imply $\tau_\phi = \tau$, how the difference $\tau_\phi - \tau$ can be bounded when balance is imperfect, and whether a representation minimizing an empirical balance criterion inherits a finite-sample causal guarantee. All additional notation is introduced only when the corresponding result requires it.

3. Identification Theory

3.1. Causal and Proxy Assumptions

The canonical results use the following assumptions, separated by their logical roles. Consistency states $Y = Y(A)$, and latent conditional exchangeability states

$$Y(a) \perp\!\!\!\perp A \mid (X, U), \quad a \in \{0, 1\}. \quad (6)$$

The pretreatment proxy channel is treatment-invariant:

$$W(a) = W, \quad W \perp\!\!\!\perp A \mid (X, U). \quad (7)$$

For each x , let $K_x(\cdot \mid u)$ denote the conditional law of W given $X = x$ and $U = u$. Conditional mixture injectivity means that equality of two mixtures $\int K_x(\cdot \mid u) dQ(u)$ implies equality of their admissible mixing laws Q . This condition is used only to interpret exact balance as full latent balance.

Every retained stratum must have positive mass and both treatment arms. Specifically, for constants $p_{\min} > 0$ and $\eta \in (0, 1/2)$,

$$P(Z = z) \geq p_{\min}, \quad \eta \leq P(A = 1 \mid Z = z) \leq 1 - \eta. \quad (8)$$

The balance metric uses a bounded characteristic kernel k on the joint space of (X, W) . Approximate results additionally require the outcome-relevant stability condition introduced below. This stability condition is not implied by injectivity and is not generally identified from one proxy.

3.2. Observable Joint Balance

For a fixed representation and retained stratum z , let

$$R_{a,z} \triangleq \mathcal{L}(X, W \mid A = a, Z = z), \quad p_z \triangleq P(Z = z). \quad (9)$$

The stratum discrepancy and its aggregate are

$$\Delta_z(\phi) \triangleq \text{MMD}_k(R_{1,z}, R_{0,z}), \quad (10)$$

$$D(\phi) \triangleq \sum_{z:p_z>0} p_z \Delta_z(\phi). \quad (11)$$

Because k is characteristic,

$$D(\phi) = 0 \iff (X, W) \perp\!\!\!\perp A \mid Z. \quad (12)$$

Exact balance is therefore an observable population property of a constructed representation. It is not assumed to be attainable.

3.3. Exact Bias Decomposition

Let $Q_{a,z} \triangleq \mathcal{L}(X, U \mid A = a, Z = z)$, let $\nu_{\phi,z} \triangleq Q_{1,z} - Q_{0,z}$, and define the latent outcome regression $\mu_a(x, u) \triangleq \mathbb{E}\{Y(a) \mid X = x, U = u\}$. If $\pi_z \triangleq P(A = 1 \mid Z = z)$, then the population bias has the exact decomposition

$$\tau_\phi - \tau = \sum_{z=1}^M p_z \int \{(1 - \pi_z)\mu_1(x, u) + \pi_z\mu_0(x, u)\} d\nu_{\phi,z}(x, u). \quad (13)$$

Equation (13) identifies the object that proxy balance must control: outcome-relevant functionals of the latent treatment-arm contrast, rather than every latent direction.

3.4. Approximate Balance and the Stability Certificate

The proxy channel sends a latent signed contrast ν to an observable kernel embedding. Denote this embedding operator in stratum z by $J_{\phi,z}\nu$. For treatment level a , denote the corresponding outcome functional by

$$\ell_{a,\phi,z}(\nu) \triangleq \int \mu_a(x, u) d\nu(x, u). \quad (14)$$

Outcome-relevant stability requires a common finite constant Γ such that, over the prespecified representation class and admissible latent contrasts,

$$|\ell_{a,\phi,z}(\nu)| \leq \Gamma \|J_{\phi,z}\nu\|_{\mathcal{H}_k}. \quad (15)$$

Combining (13) with (15) yields the established population certificate

$$\boxed{|\tau_\phi - \tau| \leq \Gamma D(\phi)}. \quad (16)$$

The constant Γ is a sensitivity parameter unless outside information calibrates it. A global RKHS outcome bridge with norm at most Γ is a sufficient structural witness for uniform stability, but the canonical method does not estimate that bridge.

3.5. Exact Identification and Attainability

If stability holds and $D(\phi) = 0$, then (16) gives $\tau_\phi = \tau$. Under proxy-channel invariance and conditional mixture injectivity, exact joint balance also implies $U \perp\!\!\!\perp A \mid X$; together with $X \perp\!\!\!\perp A \mid Z$, latent exchangeability, consistency, and overlap, this yields

$$Y(a) \perp\!\!\!\perp A \mid Z \quad (17)$$

and identifies the ATE by adjustment for Z .

Identification at an attained endpoint must be separated from existence of that endpoint. For the class of all maps from X to at most M strata, primitive strict overlap implies the established equivalence

$$\exists \phi : D(\phi) = 0 \iff W \perp\!\!\!\perp A \mid X \text{ and } e(X) \text{ takes at most } M \text{ values,} \quad (18)$$

where $e(X) \triangleq P(A = 1 \mid X)$. Thus exact balance is a strong certification result, while the practically interesting confounded regime is generally approximate.

3.6. When Discrepancy Orders Realized Bias

The bound in (16) does not generally imply that a smaller discrepancy produces a smaller realized bias. Under the established restricted alignment result with binary severity, an x -homogeneous informative proxy channel, an additive homogeneous outcome, within-stratum X balance, and a common direction of latent selection, the proxy-marginal discrepancy satisfies

$$|\tau_\phi - \tau| = \kappa D_W(\phi) \quad (19)$$

for a representation-independent $\kappa > 0$. This identity gives order preservation inside that restricted model class. Outside it, the paper claims minimization of a certified worst-case bias, not automatic minimization of realized bias.

3.7. Independent Audit-View Extension

A proxy-augmented extension splits a high-dimensional measurement into a representation view and an untouched audit view, writes

$$W = (W^{\text{rep}}, W^{\text{audit}}), \quad R = \rho(X, W^{\text{rep}}), \quad (20)$$

and evaluates $(X, W^{\text{audit}}) \perp\!\!\!\perp A \mid R$. This prevents the representation from making audit balance tautological by copying the audited coordinate. Exact point identification is available only under additional multi-view conditional independence and an outcome-relevant null-space or injectivity condition for the audit channel. This extension is currently a conditional theorem direction, not

part of the canonical established theory. Its approximate stability bound is also open and will not be presented as a proved result.

4. Estimating Causal Effects

4.1. Outcome-Free Representation Learning

For a candidate map ϕ , let $\hat{\mu}_{a,z}$ be the empirical kernel mean embedding of (X, W) among observations with $A = a$ and $\phi(X) = z$. Define

$$\hat{\Delta}_z(\phi) \triangleq \|\hat{\mu}_{1,z} - \hat{\mu}_{0,z}\|_{\mathcal{H}_k}, \quad (21)$$

$$\hat{D}(\phi) \triangleq \sum_z \hat{p}_z(\phi) \hat{\Delta}_z(\phi). \quad (22)$$

The ideal empirical learner returns $\hat{\phi}$ satisfying

$$\hat{D}(\hat{\phi}) \leq \inf_{\phi \in \Phi} \hat{D}(\phi) + \varepsilon_{\text{opt}}, \quad (23)$$

where ε_{opt} records optimization error. Neither Y nor U enters this training objective. Minimum stratum mass and representation-level overlap are enforced as admissibility conditions rather than inferred from a small penalty alone.

4.2. Cross-Fitted Algorithm

The operational algorithm has four stages. First, split the sample into representation-training and effect-estimation folds. Second, learn $\hat{\phi}$ on the training fold by minimizing an outcome-free balance objective with Adam or a finite-class search. Third, evaluate overlap and the balance discrepancy on data not used to choose the representation. Fourth, set $Z = \hat{\phi}(X)$ on the effect-estimation fold and estimate the adjusted contrast. Rotating the folds and averaging estimates uses every observation while preserving the conditional fixed-representation argument.

Current neural implementations optimize differentiable soft assignments and may deploy a hard map. If $\hat{D}_{\text{soft}}$ is used in place of the analyzed hard-stratum objective, a valid theorem needs an explicit soft-to-hard error $\varepsilon_{\text{soft}}$. The existence of such a uniform bound for the current networks is open; the manuscript will report the training surrogate and held-out hard discrepancy separately.

4.3. AIPW Estimation for a Fixed Representation

For fixed finite Z , define the nuisance functions only now:

$$e_Z(z) \triangleq P(A = 1 \mid Z = z), \quad (24)$$

$$m_a(z) \triangleq \mathbb{E}(Y \mid A = a, Z = z). \quad (25)$$

Given cross-fitted estimates $\hat{e}_Z$ and $\hat{m}_a$, the augmented inverse-probability weighted estimator is

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{m}_1(Z_i) - \hat{m}_0(Z_i) + \frac{A_i \{Y_i - \hat{m}_1(Z_i)\}}{\hat{e}_Z(Z_i)} \right. \quad (26)$$

$$\left. - \frac{(1 - A_i) \{Y_i - \hat{m}_0(Z_i)\}}{1 - \hat{e}_Z(Z_i)} \right]. \quad (27)$$

For a fixed finite representation, the corresponding influence function is

$$\psi_\phi(O) = \frac{A\{Y - m_1(Z)\}}{e_Z(Z)} - \frac{(1 - A)\{Y - m_0(Z)\}}{1 - e_Z(Z)} + m_1(Z) - m_0(Z) - \tau_\phi. \quad (28)$$

The stratified plug-in estimator equals (27) when empirical cell means are used, and it is asymptotically efficient for τ_ϕ in the nonparametric model for (Z, A, Y) . This statement is established for fixed ϕ ; a complete joint limit theory for a neural, data-selected $\hat{\phi}$ remains open.

4.4. Finite-Sample Error Decomposition

Suppose the representation class is finite and, with probability at least $1 - \delta$,

$$\sup_{\phi \in \Phi} |\hat{D}(\phi) - D(\phi)| \leq r_n(\delta). \quad (29)$$

For a bounded kernel and the overlap and mass constants in Section 3, the established finite-class analysis gives

$$r_n(\delta) = O\left(\sqrt{\frac{\log |\Phi| + \log(1/\delta)}{np_{\min}\eta}} + \sqrt{\frac{M + \log |\Phi| + \log(1/\delta)}{n}}\right), \quad (30)$$

with explicit constants available from the count, kernel-embedding, and multinomial-weight concentration steps. Combining (23) with the population certificate yields

$$|\tau_\phi - \tau| \leq \Gamma\{\hat{D}(\hat{\phi}) + r_n(\delta)\}, \quad (31)$$

$$|\tau_\phi - \tau| \leq \Gamma\left\{\inf_{\phi \in \Phi} D(\phi) + 2r_n(\delta) + \varepsilon_{\text{opt}}\right\}. \quad (32)$$

If the effect estimator also satisfies $|\hat{\tau} - \tau_\phi| \leq \varepsilon_{\text{est}}$, then

$$\boxed{|\hat{\tau} - \tau| \leq \varepsilon_{\text{est}} + \Gamma\{\hat{D}(\hat{\phi}) + r_n(\delta)\}.} \quad (33)$$

For bounded outcomes and a fixed finite representation, ε_{est} has an explicit Hoeffding and multinomial bound. Equation (33) is therefore established for enumerable classes and fixed-fold evaluation. A neural-class complexity bound, a proved $\varepsilon_{\text{soft}}$, and joint inference after adaptive representation selection are explicitly open. If these pieces are supplied, the right side of (32) gains the corresponding optimization, discretization, and nuisance-estimation terms rather than changing the population target.

5. Experiments

5.1. Questions, Comparators, and Reporting Rules

The experiments answer four separate questions. First, does proxy-supervised representation adjustment reduce ATE error relative to adjustment using X alone? Second, do the exact-assumption designs recover the known balancing structure? Third, does the finite-state experiment verify the established certificate and its failure as the proxy channel becomes ill-posed? Fourth, does the method show a useful descriptive signal in an observational benchmark with a noisy randomized reference? We report mean absolute error, root mean squared error, paired wins, held-out discrepancy, overlap diagnostics, and Monte Carlo uncertainty. Oracle procedures that observe U or the true balancing score are simulation references, not implementable competitors.

5.2. Honest Approximate SCMs

The primary approximate study fixes the causal order $U \rightarrow (X, W)$, $(X, U) \rightarrow A$, and $(A, X, U) \rightarrow Y$. Both SCMs use $\dim(X) = 20$ and $\dim(W) = 10$, with twenty independently generated data sets and nested prefixes $n \in \{100, 200, 500, 1000, 2000, 5000, 10000\}$. SCM1 uses linear proxy measurements and a nonlinear outcome component; SCM2 changes the structural equations to nonlinear transformations while preserving the same causal order and an informative monotone proxy channel. The proposed cross-mask method learns a representation from X and all but one proxy coordinate, audits balance with the held-out coordinate, and averages over masks.

At $n = 1000$, the mean absolute errors of proposed/ X -adjustment/full- (X, W) /oracle- (X, U) were 0.169/0.601/0.310/0.130 in SCM1 and 0.112/0.440/0.188/0.107 in SCM2. At $n = 10000$, the corresponding proposed errors were 0.199 and 0.134, versus 0.588 and 0.413 for X adjustment. These are finite-sample simulation results, not a proof of the audit-view identification theorem. The self-conditioned heuristic remained substantially biased, which is consistent with the self-conditioning warning in Section 3.

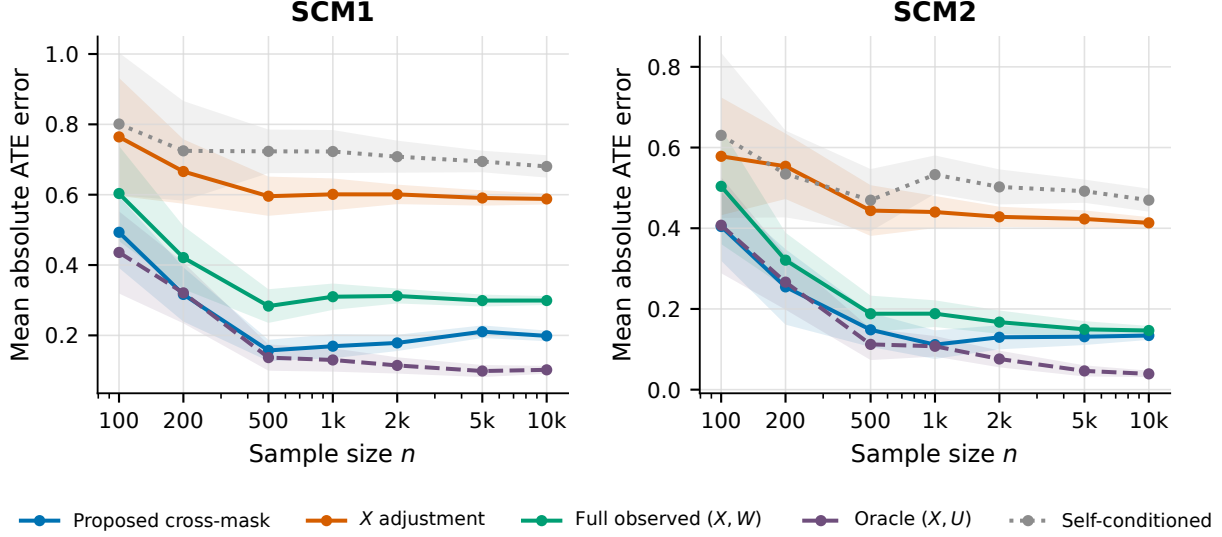


Figure 1: Honest approximate SCMs. Curves show mean absolute ATE error across twenty independently generated data sets; bands are approximate 95% Monte Carlo intervals for the mean. Prefixes are nested within a data set, so uncertainty is computed separately at each n and is not pooled across sample sizes. The proposed cross-mask learner uses one proxy coordinate for held-out auditing and averages across masks.

5.3. Exact-Assumption Tracks A and B

Track A uses a finite balancing score, $\dim(X) = 20$, nine representation-side proxy coordinates, and one independent audit coordinate. The DGP enforces strict population overlap, exact decoding of the discrete confounding state from the representation-side view, and an injective Gaussian audit channel. At $n = 10000$ over twenty replications, mean absolute error was 0.175 for the learned audit representation, 0.433 for X adjustment, 0.186 for direct full-observed adjustment, and 0.033 for the oracle balancing score. The primary comparison with X passed, while the prespecified secondary superiority criterion against full-observed adjustment failed.

Track B uses a continuous exact balancing score and a linear representation class matched to that score. At $n = 10000$, mean absolute error was 0.0748 for the learned representation, 0.1583 for fitted X adjustment, and 0.1342 for fitted full-observed adjustment. The learned method beat X in 20/20 and full-observed adjustment in 19/20 paired replications. Its mean rank-recovery Spearman correlation was 0.915. The held-out conditional-dependence diagnostic used finite random features and is a surrogate, not an exact population KCI test; the design also covers one prespecified scalar- U SCM family.

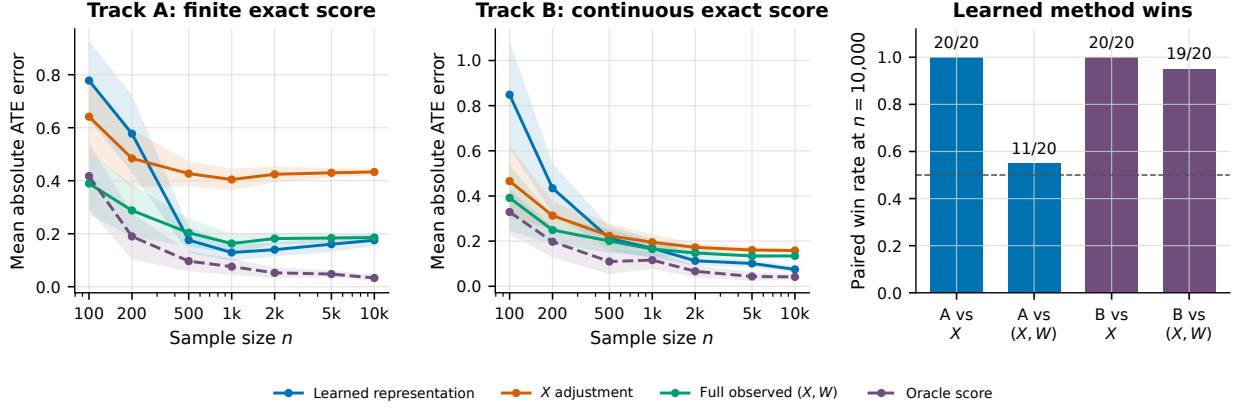


Figure 2: Exact-assumption experiments. The first two panels report mean absolute ATE error with approximate 95% Monte Carlo intervals over twenty replications. The third panel reports paired win rates at $n = 10,000$, where a win means that the learned representation has smaller absolute error than the indicated comparator. Track A satisfies the primary comparison with X but not the prespecified secondary superiority criterion against full observed adjustment. Track B records rank-recovery Spearman correlation 0.915 and held-out finite-RFF conditional-dependence surrogate 0.00691.

5.4. Finite-State Verification of the Certificate

The population verification enumerates all 3^{10} candidate maps, of which 55,980 satisfy the admissibility conditions. Exactly $3! = 6$ maps attain exact balance, and their biases are zero to machine precision. The common certificate has zero violations over all admissible maps. As the proxy channel approaches non-injectivity, the relevant stability constant grows from 1.64 to 48.9 and then diverges, even while the observable discrepancy can shrink. This is the intended sensitivity behavior: a small discrepancy is informative only together with a credible stability budget. The explicit finite-sample radius is valid but conservative in this design.

5.5. WSC Observational Benchmark

The WSC study forms an observational subset of 1,105 participants whose randomized mathematics assignment agrees with pretreatment preference, and targets the population ATE in the original 2,200 randomized participants. The full randomized difference in means is 0.760 with standard error 0.151 and serves only as a noisy benchmark. Across ten algorithmic split repetitions on this one fixed data set, the unconstrained cross-mask estimate averaged 1.302 with split standard deviation 0.065 and absolute benchmark gap 0.541, compared with gaps 2.690 for the naive observational contrast, 1.704 for direct X adjustment, and 0.672 for direct (X, W) adjustment. The unconstrained cross-mask method is the proposed WSC method shown in Figure 3. Its practical-overlap gate passed 0/10 repetitions. The benchmark therefore supplies descriptive evidence only, not a validated causal-performance claim, causal ground truth, or a test of proxy validity.

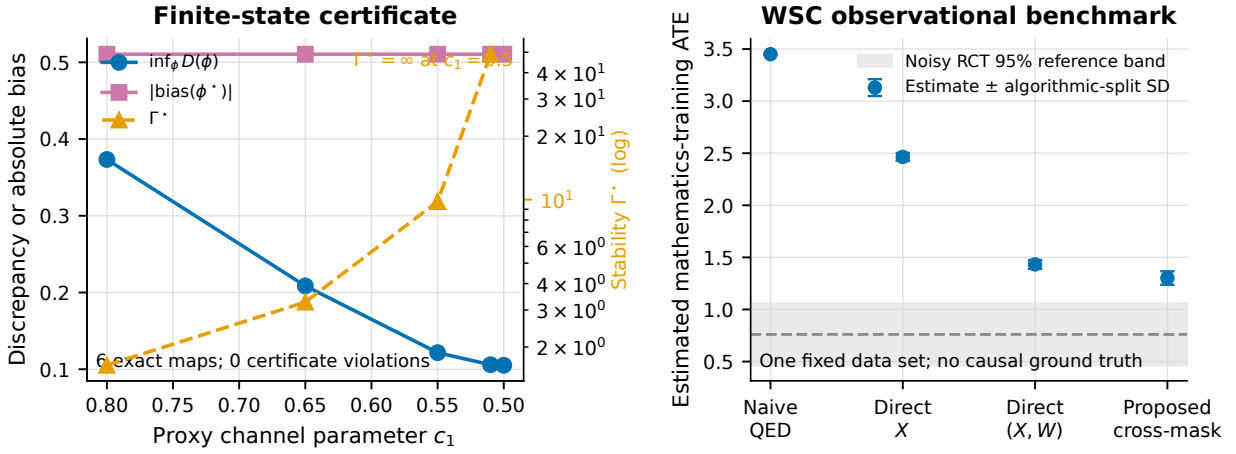


Figure 3: Certificate and observational diagnostics. Left: as the finite-state proxy channel approaches non-injectivity, the observable discrepancy decreases while realized bias remains fixed and the required stability constant diverges. Right: WSC estimates relative to the noisy randomized reference. Error bars for fitted WSC methods are standard deviations across ten algorithmic splits of one fixed data set, not Monte Carlo uncertainty. The WSC panel has no causal ground truth and reports only the unconstrained cross-mask learner as the proposed method.

6. Discussion and Limitations

6.1. What the Method Delivers

The canonical method uses a single pretreatment proxy in two roles that are available from observed data: supervising an outcome-free representation and measuring the residual imbalance of (X, W) across treatment arms within that representation. Under uniform outcome-relevant stability, the product $\Gamma D(\phi)$ is a sensitivity certificate for hidden-confounding bias. This is stronger than reporting a conditional-independence test alone because it targets a causal error scale, but weaker than point identification when $D(\phi) > 0$ because Γ is not generally identified from one proxy.

6.2. Testable and Structural Components

Held-out joint balance, empirical overlap, and minimum stratum mass are observable properties. Proxy-channel invariance, conditional mixture injectivity, and outcome-relevant stability are structural assumptions about the full-data law. Exact balance is therefore a testable endpoint only up to sampling error, while its causal interpretation still requires structural assumptions. A large held-out discrepancy can falsify exact balance; a small test statistic cannot prove exact balance, injectivity, or stability.

6.3. Attainability and Realized Bias

For $Z = \phi(X)$ with finitely many strata, exact balance is attainable only on a narrow boundary characterized in Section 3. This is a strength as a safety certificate and a limitation as a universal identification method. In the approximate regime, reducing $D(\phi)$ reduces the certified worst-case bias under a common Γ , but it need not reduce the realized bias without additional alignment conditions. Cancellation, outcome-irrelevant proxy directions, bias floors, and ill-posed channels explain why the two orderings can differ.

6.4. Learning and Inference Gaps

The current finite-sample theorem covers enumerable representation classes and fixed-fold evaluation. The neural experiments use soft assignments and covariance-weighted surrogate objectives that are not algebraically identical to the canonical MMD aggregation. A neural-class generalization bound, a soft-to-hard guarantee, and valid joint inference after adaptive representation selection remain open. These gaps prevent the manuscript from describing every neural result as a direct verification of the theory, even when its empirical error is lower.

6.5. Audit Views and Practical Scope

Cross-mask learning uses one proxy coordinate for audit and the remaining high-dimensional measurement for representation learning, then averages across audit orientations. This design blocks literal self-copying of the audited coordinate. It does not by itself establish conditional independence among proxy views, so point identification still needs a multi-view structural assumption and

an audit-channel null-space condition. The WSC experiment further shows that an encouraging benchmark gap can coexist with a failed practical-overlap rule. Real-data evidence must therefore report overlap, audit discrepancies, and benchmark uncertainty without treating any one diagnostic as ground truth.

6.6. Scope of the Contribution

The defensible contribution is the combination of a proxy-supervised balancing representation, an observable discrepancy, an exact certification boundary, and a stability-indexed approximate guarantee. The paper does not claim that one generic proxy point-identifies the ATE, that every smaller discrepancy has smaller realized bias, or that the current neural optimizer reaches the population optimum. A systematic literature audit and broader external validation are required before any firstness or universality claim.

7. Conclusion

Single proxy balancing asks a deliberately limited question: what can an observed pretreatment proxy reveal about the hidden-confounding bias that remains after learning a low-dimensional adjustment representation? For the canonical construction $Z = \phi(X)$, exact observable joint balance identifies the ATE under the stated proxy conditions, while approximate balance yields the established sensitivity certificate $|\tau_\phi - \tau| \leq \Gamma D(\phi)$. Empirical minimization of the discrepancy selects the smallest certified worst-case bias in the prespecified class up to generalization and optimization error. The current simulations support this mechanism in exact and approximate designs and expose important failures, including self-conditioning, ill-posed proxy channels, surrogate mismatch, and practical-overlap loss. Completing the neural learning theory and independently validating the audit-view extension are the principal steps needed to turn the present manuscript skeleton into a submission-ready paper.

References

- Cui, Y., Pu, H., Shi, X., Miao, W., and Tchetgen Tchetgen, E. (2024). Semiparametric proximal causal inference. *Journal of the American Statistical Association*, 119(546):1348–1359.
- Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., and Smola, A. (2012). A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773.
- Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, 20(1):25–46.
- Imai, K. and Ratkovic, M. (2014). Covariate balancing propensity score. *Journal of the Royal Statistical Society: Series B*, 76(1):243–263.
- Lipsitch, M., Tchetgen Tchetgen, E., and Cohen, T. (2010). Negative controls: A tool for detecting confounding and bias in observational studies. *Epidemiology*, 21(3):383–388.
- Miao, W., Geng, Z., and Tchetgen Tchetgen, E. J. (2018). Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika*, 105(4):987–993.
- Park, C., Richardson, D. B., and Tchetgen Tchetgen, E. J. (2024). Single proxy control. *Biometrics*, 80(2):ujae027.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2 edition.
- Rosenbaum, P. R. and Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55.
- Shalit, U., Johansson, F. D., and Sontag, D. (2017). Estimating individual treatment effect: Generalization bounds and algorithms. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 3076–3085.
- Tchetgen Tchetgen, E. J., Ying, A., Cui, Y., Shi, X., and Miao, W. (2024). An introduction to proximal causal inference. *Statistical Science*, 39(3):375–390.

A. Simulation Details

The existing project-local experiments will be documented by recording each data-generating process, parameter value, sample size, replication count, and random seed policy. Every manuscript figure and table will be linked to the script and machine-readable result from which it was produced.

The simulation audit will separate observed quantities from oracle quantities. In particular, any result that uses U or a direction constructed from U will be labeled as an oracle diagnostic and will not be presented as the output of a learnable procedure.

B. Proof Obligations

The first proof obligation is a transfer result from an observable joint balance condition to a latent balance condition. Its assumptions must state the proxy measurement relation and the injectivity or completeness property used in the transfer.

The second proof obligation is identification of τ from the observed data distribution after latent balance has been established. The proof must list every causal assumption and show where each assumption enters the identifying functional.

The third proof obligation is a boundary construction. It must exhibit two latent data-generating processes that agree on the relevant observed information but disagree on the target whenever a required transfer assumption is removed.